# INVENTIX - ANTIGRAVITY GENERATION PROMPT

**How to use this document:** Copy and paste the prompt in the dark box below directly into Antigravity or your AI code generator to scaffold the full Inventix web application according to the specifications.

## MASTER DEVELOPER PROMPT

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You are an expert Lead Software Engineer building "Inventix", a production-ready Web Event Management System tailored for the Cameroonian market using Django (Python), SQL Database, and modern Web Tools. ### PROJECT REQUIREMENTS & TECH STACK - Framework: Django 4.x (Python) with MVT Architecture. - Database: PostgreSQL or SQL-Server (Relational models with Django ORM). - Frontend: Responsive HTML5, CSS3 (Tailwind CSS via CDN), and vanilla JavaScript. - Architecture: 3-Tier Client-Server model with strict RBAC. ### CORE ACTORS & ROLES 1. Visitor/Guest: Browse event catalog, search by category/city (Yaoundé, Douala), view details. 2. Attendee: Register account, book seats, view personal dashboard, download/view QR tickets. 3. Organizer: Create/manage events, configure maximum seat capacities, view real-time attendee roster, scan/verify door QR pass. 4. Administrator: Django admin access for user roles, event moderation, and data integrity. ### EVENT TYPES TO SUPPORT - Academic/Professional (Conferences, Seminars) - Entertainment (Concerts, Festivals) - Sports (Football/Basketball Matches) - Private Events (Marriages, Birthdays with strict RSVP capacity locking) ### REQUIRED DJANGO MODELS (models.py) 1. CustomUser (AbstractUser): role choices = ['ATTENDEE', 'ORGANIZER', 'ADMIN'] 2. Category: name, slug, description 3. Event: organizer (FK CustomUser), title, category (FK Category), event_type, description, venue_address, city, start_datetime, end_datetime, max_capacity, remaining_seats, price, status 4. Booking: attendee (FK CustomUser), event (FK Event), ticket_code (UUID/QR Token), seats_booked, booking_date, status (CONFIRMED, CANCELLED, CHECKED_IN) ### CORE FUNCTIONAL FEATURES TO IMPLEMENT 1. User Auth: Custom registration with role selection, login, logout, and login_required decorators. 2. Public Event Discovery: Search bar + category filters + city dropdown. 3. Atomic Booking System: View function with @transaction.atomic ensuring seats do not drop below zero on high concurrency. 4. Organizer Dashboard: Event listing, attendee count, seat percentage progress bars, and CSV attendee exporter. 5. Door Scanner UI: Simple Javascript/HTML web camera or manual token validator for check-in verification. Generate clean, modular, and well-commented code for the core Django App (models.py, views.py, urls.py, forms.py, and main HTML templates).
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